CASPA SCHOOL OS

Complete School Management, Administration,
Academic, Safety & AI Platform

UI/UX DESIGN GUIDELINES

Pre-Primary to Grade 12 • Web + Mobile • Role-Based Enterprise Platform

Product Vision	One School. One Platform. Complete Visibility. Controlled Access. Intelligent Automation.
Primary Users	Management, Principal, Admin, Teachers, Students, Parents, HR, Accounts, Admission, Transport, Hostel, Safety, Library, Alumni and authorized staff.
AI Layer	CASPA — an AI learning, teaching, school-intelligence and decision-support assistant.

Prepared as a starting blueprint for the UI/UX team. The document defines product hierarchy, screens, role experiences, design system direction, workflows, dashboards and safety-sensitive UX requirements.

1. Product Vision & UX North Star

CASPA School OS is a unified School Operating System for institutions serving Pre-Primary through Grade 12. It combines school ERP, student information, academics, admissions CRM, HRMS, payroll, accounts, fees, transport, hostel, library, safety, communication, alumni and AI.

UX North Star: Every user should see a simple, role-specific workspace that answers: What is happening? What needs attention? What can I do next?

Core experience principles

- Role-first: the interface changes according to the user's responsibilities.
- Permission-first: visibility and actions are controlled at module, record and action level.
- Action-oriented dashboards: KPIs must lead to actionable drill-downs.
- Age-appropriate student UX: Pre-Primary and Primary experiences should be simpler than Grades 9–12.
- Sensitive-by-default: safeguarding, counselling, HR and financial information require restricted access.
- Mobile-first for parents, students and teachers; desktop-first for administration and operations.
- Progressive disclosure: do not expose 300 features at once. Reveal complexity only when needed.

2. Product Ecosystem / Information Architecture

Domain	Key Areas
People	Students, Parents, Teachers, Employees, Alumni, Guardians
Academics	Curriculum, Classes, Subjects, Syllabus, Timetable, Attendance, Homework, Assignments, Exams, Results
Business	Admissions CRM, Fees, Accounts, Payroll, Procurement, HRMS
Operations	Transport, Hostel, Library, Inventory, Assets, Events
Safety	Anti-Ragging, Child Protection, Counselling, Discipline, Grievance, Emergency, Visitor/Security
Engagement	Communication, PTM, Notices, Activities, Clubs, Sports, Alumni
Intelligence	Analytics, Reports, CASPA AI, School Intelligence
Platform	RBAC, Workflow Engine, Audit Logs, Integrations, Configuration, Multi-Campus

3. Role & Workspace Architecture

Do not build one universal dashboard. Create role-specific workspaces and allow users with multiple responsibilities to switch workspaces.

Role	Primary Workspace
Super Admin / Management	Organization, schools, campuses, configuration, global reports, permissions, integrations
Principal	School command center, approvals, academics, staff, finance summaries, safety, operations

Vice Principal / Academic Head	Academic performance, teachers, timetable, lesson plans, examinations
School Admin	Students, parents, classes, documents, certificates, communications, operations
Teacher	Classes, attendance, lesson plans, homework, assignments, marks, student support
Student	Learning, timetable, assignments, exams, results, activities, CASPA
Parent	Children, progress, attendance, fees, notices, transport, PTM, certificates
HR	Employees, recruitment, attendance, leave, payroll, performance, exit
Accounts	Fees, receipts, dues, expenses, payroll, budgets, reports
Admission Team	Leads, counselling, applications, tests, interviews, conversion
Transport Manager	Buses, routes, drivers, live tracking, maintenance, student allocation
Warden	Hostel rooms, attendance, movement, visitors, incidents
Safety / Anti-Ragging Officer	Confidential incidents, investigation, escalation, follow-up
Counsellor	Appointments, referrals, restricted counselling records
Librarian	Books, issue/return, fines, reservations
Security	Visitors, gate entry, pickup verification, emergency workflows
Custom Role	Admin-defined combination of modules, records and actions

4. Role-Based Access & Custom Permission Engine

This is a foundational UX requirement. Admin must be able to create or modify roles without developer involvement.

- Module permissions: View, Create, Edit, Delete.
- Action permissions: Approve, Reject, Publish, Export, Download, Print, Share, Generate.
- Record scope: Own records, Assigned records, Department, Class/Section, Campus, School, Organization.
- Time-bound permissions: start date, expiry date and automatic revocation.
- Data-field restrictions: e.g., teacher can see academic data but not fee information.
- Custom role builder with permission search, grouped modules, selected count and permission preview.
- Permission change confirmation for sensitive roles and automatic audit logging.

UI requirement: show a clear 'Why can I see this?' / 'Access controlled' indicator where appropriate, especially on sensitive records.

5. Web Dashboard Design

Principal Command Center

- Top KPI cards: Students, Teachers/Staff, Attendance, Fee Collection, Outstanding Fees, Admissions, Safety Alerts, Transport, Hostel.
- Today's agenda: events, examinations, PTM, meetings, approvals.

- Critical alerts panel: safety incidents, transport incidents, overdue approvals, academic risk.
- School performance: attendance trend, academic trend, fee trend, admission funnel.
- Operations snapshot: active buses, hostel occupancy, library/asset issues.
- Quick actions: announcement, approval, certificate, incident review, report generation.
- CASPA command box: natural-language school intelligence queries.

Admin dashboard

- People, academics, finance, operations, safety and pending approvals.
- Global search and filters by campus, class, section, department and academic year.
- Configurable widgets with drag-and-drop layout.

6. Academic & Student Management Screens

Academic Management

- Academic year/session
- Campus, classes and sections
- Subject and elective management
- Curriculum and syllabus
- Lesson planning
- Timetable
- Homework
- Assignments
- Study material
- Learning progress
- Academic calendar

Student Information

- Student directory
- Student profile
- Parent/guardian profile
- Emergency contacts
- Documents
- Academic history
- Attendance
- Activities
- Achievements
- Certificates
- Transport allocation
- Hostel allocation
- Student portfolio

Examination

- Exam setup
- Exam schedule
- Question bank
- Question paper
- Invigilation
- Marks entry
- Internal assessment
- Practical marks
- Grade calculation
- Result publishing
- Report card
- Performance analytics

7. Student, Parent & Teacher Experiences

Student mobile navigation

Recommended bottom navigation: Home | Classes | Tasks | CASPA | Profile

- Today's timetable and next class.
- Attendance and academic progress.
- Homework/assignment inbox with due dates.
- Exam schedule and results.
- Study resources.
- Activities, achievements and certificates.
- CASPA learning assistant.

Parent mobile navigation

Recommended bottom navigation: Home | Child | Fees | Messages | More

- Child switcher for multiple children.
- Attendance, academic performance, teacher feedback and participation.
- Fees, bills, receipts and online payments.
- Notices, PTM, certificates and NOC.
- Transport and bus status.
- Important school messages and emergency alerts.

Teacher mobile navigation

Recommended bottom navigation: Home | Classes | Attendance | Tasks | More

- Today's classes and periods.
- One-tap attendance.
- Homework and assignment creation.
- Marks and feedback.
- Student alerts.

- Parent communication and PTM.
- CASPA teaching assistant.

8. Operations Modules

Admissions CRM

- Enquiry
- Lead pipeline
- Counselling
- Application
- Document verification
- Entrance test
- Interview
- Selection
- Fee payment
- Admission confirmation
- Conversion analytics

HRMS

- Employee directory
- Recruitment/ATS
- Onboarding
- Documents
- Attendance
- Leave
- Payroll
- Performance
- Training
- Promotion/transfer
- Exit

Accounts & Fees

- Fee structures
- Invoices
- Receipts
- Dues
- Discounts
- Scholarships
- Refunds
- Expenses
- Budgets

- Bank/cash
- Reports

Library

- Catalogue
- Copies
- Issue/return
- Renewal
- Reservation
- Fine
- Lost/damaged
- Digital resources

Inventory & Procurement

- Assets
- Stock
- Purchase request
- Approval
- Vendor
- Purchase order
- Delivery
- Verification
- Maintenance

Events / Sports / Clubs

- Event creation
- Registration
- Attendance
- Participants
- Results
- Achievements
- Certificates
- Club membership

9. Transport Management UX

- Transport command center: active buses, delayed buses, routes, drivers, student allocation, maintenance and document alerts.
- Bus profile: registration, capacity, driver, conductor, route, GPS, insurance, fitness, permit, maintenance.
- Route builder with map, stops, timings, assigned bus and student list.
- Live tracking for authorized users; parents see only the route/bus relevant to their child.
- Boarding/drop tracking using QR/RFID or supported hardware where implemented.

- Alerts for delay, route deviation, vehicle issue and document expiry.
- Driver mobile mode should be extremely simple: today's route, student list, boarding, emergency.

10. Hostel Management UX

- Hostel command center: occupancy, available beds, students, wardens, attendance, incidents and visitors.
- Building → Floor → Room → Bed hierarchy.
- Student allocation and room transfer workflow.
- Hostel attendance: present, absent, leave, outing, medical, approved movement.
- Movement management: departure, purpose, destination, expected return, actual return, authorization.
- Visitor registration with verification and entry/exit timestamps.
- Emergency and incident reporting with restricted visibility.

11. Anti-Ragging, Child Safety & Safeguarding UX

Safety modules must be treated as high-sensitivity areas. Design for confidentiality, least privilege, clear escalation, evidence integrity and auditability.

Safety reporting

- Report an Incident entry point available to permitted students, parents and staff.
- Categories: ragging, bullying, harassment, threat, intimidation, physical incident, verbal abuse, cyberbullying, discrimination, transport incident, hostel incident and other safety concerns.
- Confidential reporting: identity visible only to authorized personnel.
- Anonymous reporting may be supported where policy allows; clearly explain that anonymity can limit investigation/follow-up.
- Risk classification: Critical, High, Medium, Low.
- Immediate escalation for critical incidents according to school policy.

Case workflow

Report → Initial Review → Risk Classification → Assign Officer → Investigation → Evidence/Statements → Action → Committee/Principal Review → Resolution → Follow-up → Closure

Additional safeguarding

- Child protection concerns.
- Counselling referrals and restricted session records.
- Discipline/incident history with rehabilitation-oriented presentation.
- Grievance management.
- Emergency control center.
- Audit trail for every sensitive action.

12. Communication, PTM & Notification Center

- Central announcement composer with audience targeting.
- Target by school, campus, class, section, role, transport group, hostel group or custom group.

- Channels: in-app, push, email, SMS and WhatsApp integration where available.
- Parent-teacher messaging with appointment requests, PTM booking and meeting history.
- Notification center categorized as Academic, Finance, Attendance, Transport, Hostel, Safety, Events, Approval and System.
- Emergency messages should have distinct visual treatment and confirmation.

13. Certificates, Alumni & Career

- Certificate templates with school branding.
- Bonafide, study, transfer, character, conduct, participation, achievement, sports, event and fee certificates.
- Unique certificate number and QR verification.
- Alumni profiles, graduation year, higher education, organization, industry, location and achievements.
- Alumni mentorship, internships, jobs, guest talks and donations.
- Grades 9–12 career exploration, stream selection, competitive exam support and higher-education guidance.

14. CASPA AI — UX Requirements

CASPA should appear as a consistent AI layer, but its capabilities must change according to role and permissions.

CASPA Mode	Capabilities
Student	Explain, doubt solving, practice, tests, revision, exam preparation, competitive exam preparation
Teacher	Quiz/worksheet generation, lesson plans, question generation, performance summaries, teaching assistance
Parent	Authorized child progress summary, attendance, assignment completion and teacher feedback summary
Principal	School-wide summaries, attendance trends, academic trends, operational issues and approval summaries
HR	Authorized workforce analytics and HR summaries
Accounts	Authorized fee/finance summaries and reports

CASPA UI patterns

- Persistent CASPA entry point in role dashboards.
- Suggested prompts relevant to current page.
- Conversation history.
- Source/reference area when CASPA uses school-provided content.
- Clear indication when an answer is AI-generated.
- Teacher review before AI-generated academic material is published.
- Privacy-aware responses: never reveal unauthorized student, HR, finance or safeguarding information.

15. Analytics & Reporting UX

- KPI cards with drill-down.
- Trend charts for attendance, academic performance, fees, admissions and operations.
- Class → section → student drill-down.
- Filters: academic year, campus, class, section, subject, department, date range.
- Export permissions must be role-controlled.
- Saved reports and scheduled reports for authorized users.
- CASPA can summarize authorized analytics in natural language.

16. Workflow & Approval Center

Create a universal 'My Approvals' experience.

- Leave approval.
- Admission approval.
- Fee concession.
- Purchase request.
- Expense approval.
- Certificate approval.
- Student transfer.
- Hostel movement.
- Transport request.
- Event request.

Each workflow should show: Requester → Current Stage → Required Action → SLA/age → History → Approve/Reject/Request Changes.

17. Recommended Visual Design Direction

Recommended personality: modern education + enterprise reliability. Avoid overly childish visuals for the administration product, while allowing age-appropriate student experiences.

Primary color palette

Name	Hex	Use
Deep Navy	#123B5D	Primary navigation, headings, enterprise trust
Education Teal	#176B87	Primary actions, links, selected states
CASPA Purple	#5B3FA3	AI surfaces and AI-specific actions
Sky	#3B82F6	Informational / interactive states
Success	#16A34A	Completed, paid, healthy
Warning	#D97706	Attention, pending

Danger	#DC2626	Critical safety/errors
Surface		Backgrounds
Border	#CBD5E1	Dividers and fields
Text	#25313C	Primary body text

Color usage rules

- Do not use red as a general accent. Reserve it for errors, danger and critical safety states.
- Purple should identify CASPA consistently and should not be used for unrelated modules.
- Green means success/healthy/completed—not simply 'positive information'.
- Use neutral surfaces heavily so data remains readable.
- Maintain WCAG-compliant contrast for text and interactive elements.

18. Typography & Visual Language

- Recommended family: Inter, Manrope or a similarly legible modern sans-serif.
- Use a clear type scale: display, H1, H2, H3, body, caption and labels.
- Avoid dense paragraphs in dashboards; convert information into cards, tables and summaries.
- Use consistent numeric styling for fees, attendance, percentages and KPIs.
- Use icons with text labels for important actions; do not rely on icon-only controls for critical functions.

19. Navigation Patterns

- Desktop: collapsible left sidebar + top bar.
- Top bar: global search, school/campus selector, notifications, CASPA, profile.
- Role switcher/workspace selector for multi-role users.
- Breadcrumbs for deep administrative screens.
- Sticky page actions on long forms.
- Mobile: bottom navigation for high-frequency actions; 'More' for secondary modules.
- Use command/search patterns for power users.

20. Forms, Tables & Data Entry

- Use step-by-step forms for admissions, onboarding and complex configuration.
- Auto-save where appropriate; show draft status.
- Inline validation with clear error messages.
- Bulk import with preview, validation and downloadable error report.
- Tables must support sorting, filtering, column selection, pagination and bulk actions.
- For sensitive records, require explicit confirmation before export/download.

21. Responsive, Accessibility & Age-Based UX

- Desktop-first for administrative operations; mobile-first for parents, students and field staff.
- Design breakpoints for desktop, tablet and mobile.

- Support keyboard navigation and visible focus states.
- Use accessible labels, contrast and touch targets.
- Pre-Primary/Primary student interfaces should use simpler navigation and visual hierarchy.
- Grades 9–12 can have richer dashboards for competitive exams, career and project work.

22. Security & Privacy UX

- Sensitive modules should show restricted-access indicators.
- MFA should be supported for privileged roles.
- Session timeout and re-authentication for high-risk actions.
- Audit logs for login, data access, export, download, permission changes and approvals.
- Least-privilege access should be the default.
- Parents should only access their linked children.
- Teachers should only access assigned classes/subjects unless explicitly authorized.
- CASPA must inherit the user's permissions before retrieving school data.

23. Visual / Product References for the UI Team

Use these as inspiration for interaction patterns, not as templates to copy.

Google Material Design — Use for accessibility, component logic, responsive behavior and interaction patterns.

Microsoft Fluent 2 — Useful reference for enterprise dashboards, navigation, tables, forms and productivity workflows.

Ant Design — Useful reference for dense enterprise data tables, forms, filters, administration panels and permissions.

Atlassian Design System — Good reference for workflows, status, collaboration, approvals and enterprise information architecture.

Salesforce Lightning Design System — Useful reference for CRM-style records, dashboards, permissions and configurable business workflows.

Notion / modern productivity tools — Reference for clean information hierarchy, command/search and progressive disclosure.

Duolingo-style learning UX — Reference only for age-appropriate learning motivation and simple student interactions—not for the enterprise admin UI.

24. Initial Screen Inventory for UI/UX

1. Login / OTP / MFA / Forgot Password	2. School & Campus Selector
3. Role / Workspace Selector	4. Principal Dashboard
5. Admin Dashboard	6. Teacher Dashboard
7. Student Dashboard	8. Parent Dashboard
9. Student Directory	10. Student Profile
11. Parent Profile	12. Teacher Profile
13. Employee Profile	14. Class & Section Management
15. Subject Management	16. Curriculum / Syllabus

17. Lesson Plan	18. Timetable Builder
19. Attendance	20. Homework
21. Assignments	22. Study Material
23. Exam Management	24. Question Bank
25. Marks Entry	26. Report Card
27. Admission CRM Pipeline	28. Admission Application
29. Counselling	30. Fee Dashboard
31. Fee Invoice	32. Payment / Receipt
33. HR Dashboard	34. Recruitment
35. Employee Attendance	36. Leave Management
37. Payroll	38. Transport Dashboard
39. Bus Profile	40. Route Builder
41. Live Bus Tracking	42. Hostel Dashboard
43. Room / Bed Allocation	44. Hostel Attendance
45. Visitor Management	46. Library Dashboard
47. Book Catalogue	48. Inventory
49. Procurement	50. Certificate Generator
51. Certificate Verification	52. Events
53. Sports	54. Clubs
55. Alumni	56. Career
57. Communication Center	58. PTM
59. Complaint / Grievance	60. Anti-Ragging Report
61. Safety Incident Case	62. Child Protection
63. Counselling	64. Emergency Control Center
65. CASPA Chat	66. CASPA Student Learning
67. CASPA Teacher Assistant	68. CASPA Principal Intelligence
69. Reports & Analytics	70. Approval Center
71. Role Builder	72. Permission Matrix
73. Workflow Builder	74. Audit Logs
75. System Settings	76. Integration Hub

25. UI/UX Team Deliverables

- Design system / component library.
- Desktop web responsive designs.
- Mobile designs for Student, Parent, Teacher, Warden and Driver.
- Role-based sitemap.
- Role-permission matrix.
- User flows for critical workflows.
- High-fidelity dashboard designs.

- Prototype for core workflows.
- Empty, loading, success, error and permission-denied states.
- Accessibility review.
- Developer-ready Figma components and tokens.
- Handoff documentation with spacing, typography, states and responsive behavior.

26. Recommended Design Priority

Priority	Design Scope
P0 — Foundation	Design system, login, role/workspace, navigation, permissions, global search, notifications
P0 — Core School	Principal/Admin/Teacher/Student/Parent dashboards, student profile, academics, attendance, timetable, homework, exams
P1 — Business	Admissions CRM, fees, accounts, HRMS, payroll
P1 — Operations	Transport, hostel, library, inventory, certificates
P1 — Safety	Anti-ragging, safeguarding, grievance, emergency, security
P2 — Engagement	Events, sports, clubs, alumni, career
P2 — Intelligence	CASPA, analytics, predictive/decision-support experiences

27. Final Design Direction

The platform should feel like a modern School Operating System—not a collection of 500 screens. Complexity belongs in the architecture; simplicity belongs in the user experience.

- One platform, multiple role experiences.
- Centralized school command center.
- Controlled access at every level.
- Operational modules connected through workflows.
- Safety and safeguarding treated as first-class product areas.
- CASPA presented as an integrated intelligence layer, not a generic chatbot.
- Web optimized for management and operations; mobile optimized for high-frequency field and family interactions.
- Design for future multi-school, multi-campus expansion from the beginning.

Recommended next step: convert this guideline into a screen-by-screen sitemap, role-permission matrix and Figma-ready page structure before starting high-fidelity UI design.